Learning a dome correction for Type IV hydrogen vessel sizing from 384 shell simulations, and an audit of the high-fidelity target

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Abstract

Genetic-algorithm layup optimizers for Type IV hydrogen vessels rely on sizing models that run in about a millisecond and ignore the dome. I built a CalculiX shell model of the whole vessel with geodesic winding, solved 384 designs of an 11 L vessel, and trained a multilayer perceptron (MLP) to predict the ratio C between the maximum fiber failure index of the shell model and that of a thick-cylinder model. On this dataset the MLP reaches a cross-validated R^2 of 0.862 on $\log C$ (ridge regression: 0.796) and 0.778 on 96 held-out designs. Inside the optimizer, the corrected index stays above the 95th percentile of the shell-model index for all 17 optimized designs, but above its maximum for only 4. An audit of the target explains this gap. The maximum sits at the polar opening in 205 designs and is not mesh-converged; even the ply in which it occurs changes with the mesh. It depends on a coverage floor that keeps part of every ply below its turnaround radius and on the end-cap force, which the DOE omits (applying it raises the maximum by 27 to 54 % in four designs), and in 202 designs no helical ply can reach the polar opening at all. The 95th percentile of the shell index is mesh-converged, insensitive to the end-cap force (below 1 %), and lies within 10 % of the fast-model maximum for 369 of 384 designs. With plies stopped at their turnaround, on the 182 designs where that is possible, the MLP overfits and linear models keep only part of the signal on a target that is itself not converged. The corrected optimizer remains conservative with respect to the converged 95th percentile (margin 2 % to 78 %). The learned correction is a correction of a numerical peak, usable as a conservative margin, not evidence of a physical dome effect.

1 Introduction

Type IV vessels store hydrogen at 35 MPa to 70 MPa in a polymer liner overwrapped with filament-wound carbon/epoxy. Burst is usually governed by fiber failure, and the dome, the dome–cylinder junction and the boss region are where geometry, winding angle and thickness change fastest [1–3]. Layup optimization by genetic algorithm (GA) evaluates tens of thousands of stacks, so the structural model inside the loop has to be cheap. The optimizer studied here uses an analytical thick-cylinder model that is accurate for the cylindrical section and blind to the domes.

The initial question was whether a correction learned from a dome-resolved finite-element model can make such an optimizer aware of the dome without slowing it down or modifying its code. Section 6 shows that the target itself is unreliable.

Contributions:

- A pipeline that couples a CalculiX composite-shell model, a 384-case design of experiments (DOE), a correction model compared with linear baselines on two extrapolation regimes, and the GA, with every optimized design re-solved in CalculiX.

*University of California, Berkeley (MEng). Work done at Arts et Métiers in May–June 2026, audit in October 2026. Code and data: https://github.com/clementdumeril/copv-type4-multifidelity.

- An audit of the high-fidelity target: location of the peak, mesh study up to 64×64 elements, ablation of the ply-coverage law near the pole, and a check of the geometric consistency of the DOE.
- A re-analysis with the mesh-converged quantity and with plies stopped at their turnaround radius, and a comparison of the shell model with six published burst tests, one of which it does not reproduce.

2 Related work

Multi-fidelity methods combine cheap and expensive models to reduce the cost of optimization and uncertainty propagation [4]. Discrepancy models go back to the Bayesian calibration framework of Kennedy and O'Hagan [5]; neural networks have been used to learn the map from low- to high-fidelity outputs [6]. The present work is a simple instance, a multiplicative correction learned on scalar outputs with the low-fidelity code left unchanged. Like any such method, it is only as reliable as the high-fidelity data, which is the point examined here.

Progressive-failure finite-element models of Type IV vessels reproduce burst pressures within a few percent when the layup, dome thickness and damage law are known [1, 2, 7, 8]. Dome thickness and fiber angle follow from the winding: geodesic paths obey Clairaut's relation and tows accumulate near the turnaround radius [9]; the dome contour changes the stress state near the junction [3]. Lamina failure is assessed with interactive or mode-based criteria [10–12]. The fast model uses Lekhnitskii's solution for an anisotropic layered cylinder [13]. Open burst-test data with full layup documentation are rare; the DLR dataset of Lüders et al. [14] is one exception.

3 Models

3.1 Fast analytical model

The fast model belongs to an existing GA code that the author did not write. Each ply of the cylindrical wall is a cylindrically orthotropic layer; continuity of radial displacement and radial stress at every interface gives a linear system for the stresses under internal pressure [13]. Ply stresses are rotated into the material frame and checked with the Hashin, Tsai–Wu and Puck criteria. One evaluation takes under a millisecond. The model has no dome, no boss and no change of winding angle; in all 384 designs it places the critical point in the cylinder.

3.2 Dome-resolved shell model

The reference model covers the cylinder, both domes and the polar openings with eight-node reduced-integration shells (S8R) in CalculiX 2.23 [15]. The meridian profile is hemispherical, isotensoid-like or a variable contour. Each element carries its own composite section with local ply angles and thicknesses. Helical and transition plies follow the geodesic law $\sin\alpha(r) = r_t/r$, where $r_t = R\sin\alpha_0$ is the turnaround radius of a ply wound at α_0 on the cylinder of radius R; hoop plies keep their nominal angle (86.5 to 89.5°). Near its turnaround radius a ply's thickness is multiplied by a logistic coverage weight, which tends to a floor ε below that radius (ε is sampled between 0.20 and 0.35). A geodesic ply does not exist below r_t; the code nevertheless keeps the floor there and assigns the ply an angle of $\arcsin(\min(r_t/r, 0.999)) \approx 87°$, a numerical convention rather than a winding law. The liner is not modelled. One boss is fixed axially and the other slides, so the vessel elongates without an artificial axial clamp. The internal pressure is applied with `*DLOAD` and the analysis is linear static. The generator can apply the axial end-cap force $p\pi r_{\mathrm{boss}}^2$ on the polar edges with `*CLOAD`, but this option is off in every DOE design, so the

Table 1: Lamina properties of the carbon/epoxy used in the DOE (`code/config/material_database.json`).

E_1	$E_2 = E_3$ [GPa]	G_{12}	ν_{12}	ν_{23}	X_T	X_C	Y_T [MPa]	Y_C	S_{12}
151	9	5	0.30	0.42	2000	1500	50	200	80

Table 2: Sampled parameters of the 11 L DOE (`data/doe_11l_single_boss_cases.json`).

Parameter	Range
Inner radius R	76–96 mm (cylinder 257–550 mm)
Boss radius	21.5–34.1 mm
Dome shape	hemispherical, isotensoid-like, variable contour
Helical angle on the cylinder	12–26°
Helical / transition / hoop pairs	4–5 / 1–2 / 2–3
Total thickness	10.5–26.1 mm
Coverage floor ε	0.20–0.35
Pressure	70 MPa; 52.5 and 87.5 MPa for the pressure split

polar edges carry no axial load from the boss; its effect is quantified in Section 6. Table 1 gives the lamina properties.

3.3 Fiber failure index and correction factor

Shell stresses at the section points are assigned to plies from their radial position, averaged per ply and per element, and rotated into the fiber frame. The fiber index of a ply is $|\sigma_{11}|/X_T$ in tension and $|\sigma_{11}|/X_C$ in compression. The vessel index FI is the maximum over all (element, ply) pairs; the 95th and 99th percentiles over the same pairs are also recorded. I first used the combined Hashin/Tsai–Wu/Puck index, but its matrix and shear terms reach values in the hundreds near the polar opening at service pressure. The correction factor is

$$C = \mathrm{FI}^{\mathrm{FE}}/\mathrm{FI}^{\mathrm{fast}}, \tag{1}$$

which lies between 1.00 and 4.51 over the DOE (median 1.42).

4 Design of experiments, correction model and optimizer coupling

DOE. All designs have an internal volume of 11 L; the cylinder length follows from the radius and dome shape. Table 2 lists the sampled parameters. Case roles are fixed before training: 288 training designs, 48 in-distribution holdouts, 24 designs at the edge of the geometric ranges and 24 at untrained pressures. Every design was solved at 24×24 elements (six elements along each dome meridian).

Correction models. The target is $\log C$. Inputs are 17 design and fast-model quantities, six derived ratios (pR/t, boss-to-radius ratio, length-to-radius ratio, hoop and transition pair fractions, helical–hoop angle gap) and the one-hot dome shape, standardized with training-set statistics. Four models are compared: a constant, ordinary least squares (OLS), ridge regression ($\alpha = 10$) and an MLP with hidden layers of 24 and 12 tanh units, written in NumPy with hand-coded backpropagation and trained full-batch with Adam [16] (learning rate 0.002, L2 penalty

Table 3: R^2 on $\log C$ for the original target (`data/extended_statistical_metrics.json`). CV: mean $\pm$ standard deviation over 4 repeats of 8 folds on the training designs.

	Constant	OLS	Ridge	MLP
CV, 288 training designs	-3.910 ± 0.207	0.791 ± 0.047	0.796 ± 0.046	0.862 ± 0.029
Held-out, all 96	-4.015	0.717	0.727	0.778
in-distribution (48)	-5.900	0.770	0.787	0.789
geometric boundary (24)	-2.270	0.647	0.672	0.800
untrained pressure (24)	-4.016	0.726	0.703	0.714

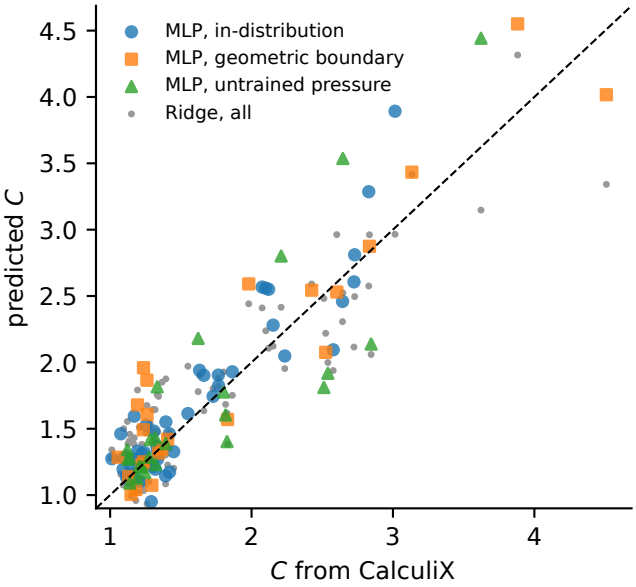


Figure 1: Predicted against CalculiX correction factor on the 96 held-out designs (`data/extended_statistical_eval_predictions.csv`).

4×10^{-4}, 6000 epochs, fixed seed). Cross-validation uses 8 folds repeated 4 times, on the training designs only. No design appears twice and none is shared between training and evaluation.

Optimizer coupling. The GA maximizes the ratio of inner to outer radius (thinnest wall), keeps the proportions of ply angles near targets, and penalizes designs whose failure index exceeds 1 at the 87 MPa design burst pressure. The coupling replaces the fast fiber index by $\mathrm{FI}^{\mathrm{fast}}\,\hat{C}(\mathbf{x})\,\exp(m_{95})$, where $\hat{C}$ is the MLP prediction and m_{95} the 95th percentile of its cross-validation residuals on $\log C$.

5 Results on the original dataset

Table 3 gives the scores on $\log C$ and Fig. 1 the held-out predictions. The MLP is ahead of ridge regression by 0.066 in cross-validated R^2, and clearly ahead only at the edge of the geometric domain; at untrained pressures OLS does slightly better. Re-running the validation script on the committed dataset reproduces every metric exactly.

The fast model always reports the cylinder as critical; the shell model reports a dome in 198 designs, the cylinder in 103, the junction in 56 and the boss edge in 27 (Fig. 2). The dome-critical designs all peak at the left pole; Section 6 shows that this is partly a labelling effect.

Ten GA runs with different seeds gave 17 distinct designs, all with 12 plies and a 15.22 mm wall, each re-solved in CalculiX at 87 MPa (Fig. 3). Without correction, the fast index is below

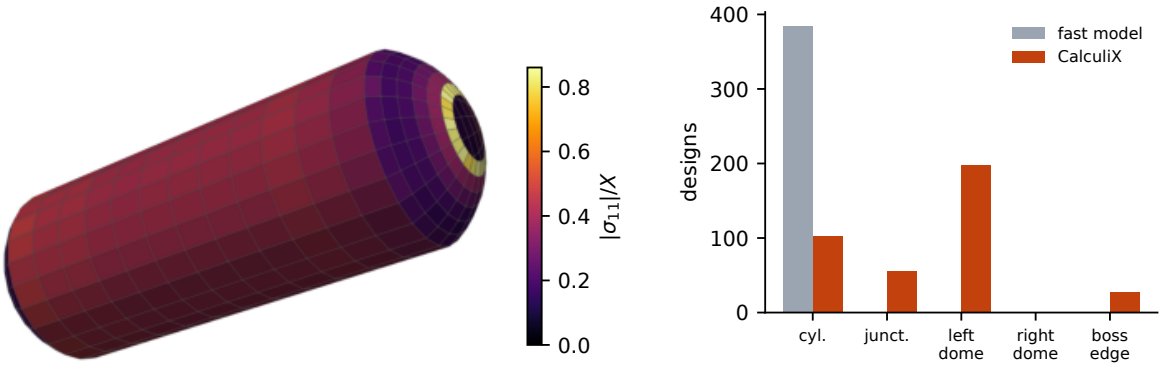


Figure 2: Left: fiber index of design `111_0122` at 70 MPa, worst ply per element. Right: zone of the maximum over the 384 designs for both models.

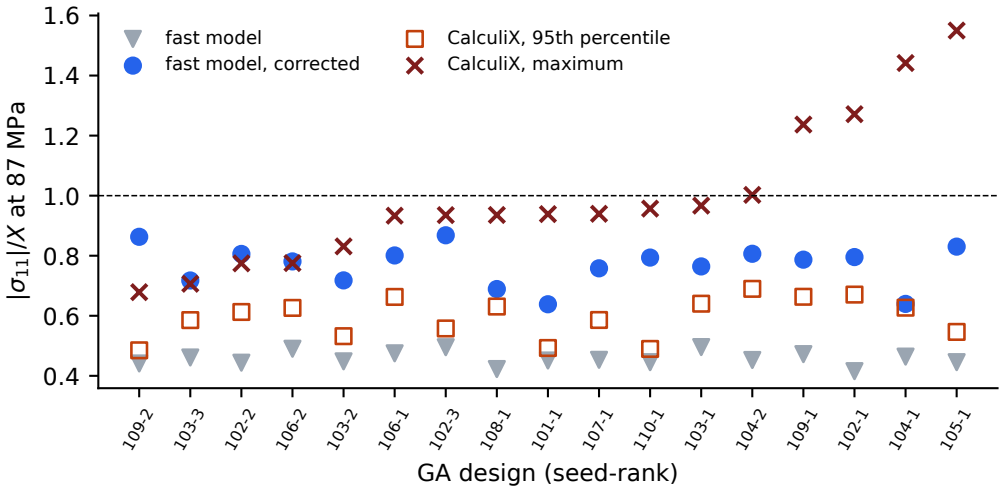


Figure 3: Fiber index of the 17 GA designs at 87 MPa (`data/ga_statistical_validation.csv`).

the CalculiX 95th percentile for all of them. With the correction, it is above the 95th percentile for 17 of 17 and above the maximum for 4; 5 designs have a CalculiX maximum above 1.

6 Audit of the high-fidelity target

6.1 Where the maximum is

In 205 of the 384 designs the maximum lies within 1.25 boss radii of the axis, that is in the ring of elements at the polar opening (`data/baseline_fiber_peak_location.csv`). In 179 of the 198 dome-critical designs it is in a hoop ply, a ply that real winding does not place in the dome. Of the 205 polar peaks, 190 are at the left pole ($z < 0$) and 15 at the right one. The generator labels the element ring next to the right polar opening "boss" and the one next to the left opening "dome"; since the DOE uses no zone-dependent material or coverage, these labels do not change the model, but they make the zone statistics asymmetric: right-pole peaks are counted as "boss edge", which is why all dome-critical designs appear at the left. The physical difference between the poles is not isolated. Re-solving the four designs of Section 6.2 with no boss ring clamped (three-point support) changes the maximum by at most 4.2 % and the 95th percentile by at most 0.8 %, with the critical zone unchanged in 4 of 4 (`data/paper_bc_minimal_pins.csv`); but the supports remain asymmetric in both variants (the two pinned nodes are on the left boss), and

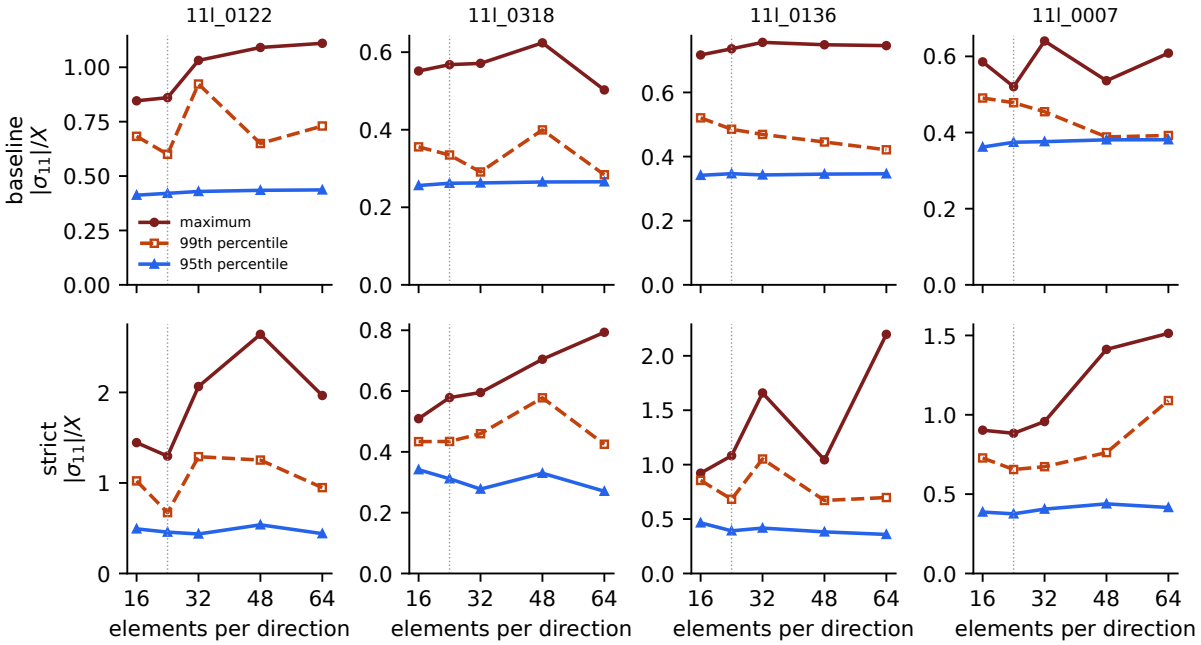


Figure 4: Fiber index against mesh density for four designs, with the DOE coverage law (top) and with plies stopped at their turnaround radius (bottom). The dotted line marks the 24×24 mesh of the DOE.

the junction element lies on the dome side at the left and on the cylinder side at the right.

The DOE also leaves the polar edges without the end-cap force $p\pi r_{\text{boss}}^2$. Applying it on both boss rings for the same four designs (`data/paper_endcap.csv`) raises the maximum by +32 to +37 % at 24×24 and by +27 to +54 % at 64×64 (design `11l_0122`: 1.15 and 1.64), and moves the critical zone of `11l_0007` from the cylinder to the dome at 24×24. The 95th percentile changes by at most 0.8 %. The maximum at the pole therefore depends on a load that the DOE omits; the 95th percentile does not.

6.2 Mesh study

Four designs were re-solved at 16, 24, 32, 48 and 64 elements per direction: three dome-critical designs close to the median C, one per dome shape, and design `11l_0007`, whose maximum is in the cylinder at 24×24 (Fig. 4, `data/paper_mesh_study.csv`; no stress sampling was applied). With the DOE coverage law (top row), the 95th percentile changes by at most 0.42 % between 48 and 64 and by at most 3.7 % between 24 and 64: it is converged at the DOE mesh. The maximum and the 99th percentile are not. They vary non-monotonically; for design `11l_0122` the maximum goes from 0.85 at 16×16 to 1.11 at 64×64, and for `11l_0007` the critical zone moves from the cylinder to the dome above 24×24. The critical ply also changes: it is a hoop ply in 4 of the 4 designs at 24×24 and a transition ply in 4 of 4 at 64×64. The "hoop ply" attribution of the DOE is therefore itself a product of the mesh.

6.3 Coverage law and geometric consistency

Removing hoop coverage from the dome does not lower the peak of design `11l_0122`: it rises from 0.86 to 1.16 and moves to a transition ply at the same radius. Any ply that the coverage floor keeps below its turnaround radius is, by the code's convention, almost circumferential there, so it carries the hoop strain of the pole whatever its thickness.

The floor also hides a geometric problem in the DOE. A helical ply wound geodesically turns around at $r_t = R \sin \alpha_0$, and to cover the polar opening without crossing it, r_t must equal the boss radius to within a band width, that is $\alpha_0 \approx \arcsin(r_{\text{boss}}/R)$. In 202 of the 384 designs

Table 4: Learning protocol on five targets (all on $\log C$). n: training/held-out designs. RMSE_0: held-out RMSE of the constant model. For ridge and MLP: CV R^2, held-out R^2 with a bootstrap 95 % interval (2000 resamples), held-out RMSE. Files: `data/extended_statistical_metrics*.json`.

			Ridge			MLP		
Target	n	RMSE_0	CV	held-out	RMSE	CV	held-out	RMSE
Maximum (original)	288/96	0.786	0.80	0.73 [0.59, 0.81]	0.183	0.86	0.78 [0.67, 0.85]	0.165
Maximum, $r_t \le r_{\mathrm{boss}}$	138/44	0.608	0.86	0.78 [0.64, 0.86]	0.164	0.78	0.70 [0.54, 0.81]	0.189
95th percentile	288/96	0.066	0.17	0.27 [0.18, 0.35]	0.046	0.27	0.48 [0.29, 0.60]	0.039
Maximum, strict	138/44	0.821	0.51	0.21 [−0.44, 0.53]	0.320	−0.10	−1.25 [−3.19, −0.29]	0.540
95th percentile, strict	138/44	0.290	0.63	0.48 [−0.01, 0.72]	0.085	0.20	−0.33 [−1.70, 0.27]	0.135

$r_t > r_{\mathrm{boss}}$: no helical ply reaches the polar opening, and the floor is what covers the pole. Among the remaining 182, 132 have $r_t < 0.9\, r_{\mathrm{boss}}$ (minimum 0.58): their helical plies would reach the edge of the opening at 35° to 64° instead of 90°, which is not windable either. Only 50 designs are close to consistent.

I re-solved the 182 designs with $r_t \le r_{\mathrm{boss}}$ with every ply stopped at its own turnaround radius ("strict"; weights below 5 % are dropped). The maximum increases (median C 2.74, up to 6.87) and stays in a dome in 177 designs, now mostly in transition and helical plies. This variant is not mesh-converged either (Fig. 4, bottom): for design `111_0122` the maximum is 1.30 at 24×24, 2.64 at 48×48 and 1.97 at 64×64, and even the 95th percentile moves by up to 18 % between 48 and 64. Dropping plies creates thickness steps that the coverage floor had smoothed out. Since most of these designs are not windable as sampled, the strict variant is a sensitivity test, not a corrected reference.

6.4 Learning other targets

Table 4 and Fig. 5 repeat the learning protocol on other targets. The 95th percentile of the shell index lies within 10 % of the fast-model maximum for 369 of 384 designs (median ratio 0.99, range 0.94–1.40). The two quantities are different statistics, and the percentile is taken over (element, ply) pairs without area weighting, so this indicates that the bulk of the shell solution is at the level predicted by the fast model rather than a pointwise agreement; where the shell maximum is in the cylinder, it is still 1.21 times the fast value at the median (up to 1.62). Little is left to learn on this target (MLP: 0.27 in CV, 0.48 held out, with a constant-model RMSE of only 0.066 on $\log C$).

Restricting the original target to the 182 designs with $r_t \le r_{\mathrm{boss}}$ (138 for training, 44 held out) keeps it learnable, so the subset alone does not explain what follows. On the strict targets the MLP overfits on 138 training designs (its hyperparameters were not re-tuned), while the linear models keep part of the signal: on the strict 95th percentile, ridge regression reaches 0.48 held out with an RMSE of 0.085 against 0.290 for the constant. That target is itself not mesh-converged, so this signal cannot be taken at face value.

7 Comparison with published burst tests

The shell model and fiber index were applied to six published vessels from four studies (Table 5); geometry and layup were rebuilt from each paper, partly from figures. The model is conservative by 4 to 22 % on five of them. For Hu et al. the pressure matches but the location does not: the first failed ply is a cylinder hoop ply in the model and in the dome in the paper. The DLR vessel is not reproduced even with the published layup book; a 22.2° helical layer becomes critical near its turnaround radius, the region that the audit identifies as unreliable.

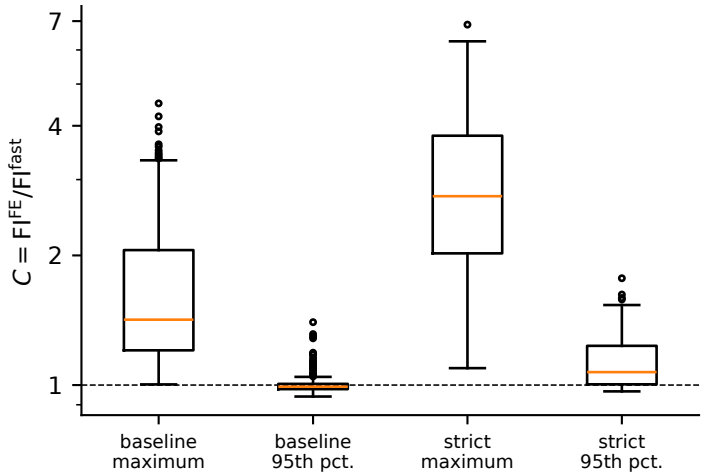


Figure 5: Distribution of the correction factor for four targets (log scale).

Table 5: Shell model against published burst tests. Ranges span the maximum, 95th and 99th percentile of the fiber index; for Jin et al. the 99th percentile at 32×32 is given.

Source	Vessel	Measured [MPa]	Model [MPa]	Gap
Hu et al. 2021 [2]	H_2, first fiber damage	161.0	148.7	−7.6 %
Agne et al. 2025 [8]	GFRP	29.39	25.4–27.7	−6 to −14 %
Agne et al. 2025 [8]	CFRP	27.04	25.7–25.9	−4 to −5 %
Jin et al. 2024 [7]	EX-A	65.20	58.6	−10 %
Jin et al. 2024 [7]	EX-B	65.17	50.8	−22 %
Lüders et al. 2025 [14]	SN03, layup book	25.37	6.5	−74 %

8 Discussion and limitations

What holds. The pipeline works end to end and is reproducible. The 95th percentile of the shell index is mesh-converged at the DOE mesh and close to the fast-model value for nearly all designs. In the optimizer, the corrected index is above that converged 95th percentile for all 17 designs, by 2 % to 78 %, whereas the uncorrected fast index is below it for all of them. On its own dataset the MLP learns C well (R^2 0.862 in CV, 0.778 held out).

What does not. The quantity the MLP learns is dominated by the maximum at the polar opening. That maximum is not mesh-converged, its critical ply changes with the mesh, it depends on a coverage floor that has no physical counterpart, it sits at a polar edge that carries no end-cap force in the DOE, and in most designs it fills a gap that the sampled winding angle could not fill. The R^2 of 0.862 measures how well a model reproduces this numerical target, not a physical dome effect, and the corrected optimizer is not conservative with respect to it. With plies stopped at their turnaround, the target becomes mesh-dependent; the MLP overfits and linear models keep only part of the signal.

Other limitations. No end-cap force on the polar edges in the DOE (its effect on four designs is given in Section 6). Linear elastic shells, no liner, no progressive damage, no residual winding stress and no liner–boss contact in the DOE (tested separately on two designs: the penalty stiffness moved stresses by about 0.3 %, tied versus penalty contact by 7.5 to 8 %, and frictional contact did not run reliably). The MLP hyperparameters were set for 288 training designs and not re-tuned for 138. No dome-only, area-weighted converged indicator was computed. One vessel family. The optimizer is not public, so the GA check cannot be re-run from the repository.

9 Conclusion and future work

A correction learned from 384 shell solutions can be coupled to a cylinder-only optimizer without slowing it down, and on its own dataset it reaches a cross-validated R^2 of 0.862. Auditing the target shows that what it corrects is mostly a mesh-dependent peak at the polar opening, produced by a coverage law, by unloaded polar edges and by helical angles that do not reach the polar opening. The next steps are a DOE that sets the helical angle from the boss radius ($\alpha_0 = \arcsin(r_{\text{boss}}/R)$ to within a band width), the end-cap force and symmetric supports at both poles,a tow-accumulation model at the turnaround instead of a coverage floor [9], a solid or locally refined model of the polar region, and a mesh-converged target checked against the measured liner contour of the DLR vessel before any further training.

Reproducibility statement

All scripts, per-design summaries and figures are in the repository. `pythoncode/calibration/extended_statistical_validation.py` reproduces Table 3 in about 3 minutes on a laptop CPU (Intel i7-6700HQ, 4 cores). The campaigns added for this paper use CalculiX 2.23 (Windows build of 15 November 2025) through `code/paper/run_campaign.py` with 2 solver threads; a 24×24 design takes about 2 minutes and a 64×64 design about 11 minutes. The raw result files (about 20 MB per design) are not versioned. `bashpaper/build_paper.sh` regenerates the figures, `paper/generated/numbers.tex` and this document.

References

[1] David Leh, Philippe Saffré, Pierre Francescato, Robert Arrieux, and Sébastien Villalonga. A progressive failure analysis of a 700-bar type IV hydrogen composite pressure vessel. *International Journal of Hydrogen Energy*, 40:13206–13214, 2015. doi: 10.1016/j.ijhydene.2015.05.061.

[2] Zhengyun Hu, Minghe Chen, and Bo Pan. Simulation and burst validation of 70 MPa type IV hydrogen storage vessel with dome reinforcement. *International Journal of Hydrogen Energy*, 46:23779–23794, 2021. doi: 10.1016/j.ijhydene.2021.04.186.

[3] Kumar C. Jois, Marcus Welsh, Thomas Gries, and Johannes Sackmann. Numerical analysis of filament wound cylindrical composite pressure vessels accounting for variable dome contour. *Journal of Composites Science*, 5:56, 2021. doi: 10.3390/jcs5020056.

[4] Benjamin Peherstorfer, Karen Willcox, and Max Gunzburger. Survey of multifidelity methods in uncertainty propagation, inference, and optimization. *SIAM Review*, 60:550–591, 2018. doi: 10.1137/16M1082469.

[5] Marc C. Kennedy and Anthony O'Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 63:425–464, 2001. doi: 10.1111/1467-9868.00294.

[6] Xuhui Meng and George Em Karniadakis. A composite neural network that learns from multi-fidelity data: Application to function approximation and inverse PDE problems. *Journal of Computational Physics*, 401:109020, 2020. doi: 10.1016/j.jcp.2019.109020.

[7] Shan Jin, Peng Cheng, Yong Bai, Jeom Kee Paik, and Jun Li. Progressive failure analysis and burst mode study of type IV composite vessels. *Ships and Offshore Structures*, 19(2): 223–232, 2024. doi: 10.1080/17445302.2022.2157192. Published online 16 December 2022.

[8] Lucas L. Agne, José Humberto S. Almeida Jr, Martin H. M. Maruyama, Diego T. de Souza, Sandro C. Amico, and Maik Tonatto. Progressive failure modelling of type IV composite overwrapped pressure vessels for compressed natural gas (CNG) storage. *Composite Structures*, 370:119424, 2025. doi: 10.1016/j.compstruct.2025.119424.

[9] Hui Wang, Shuang Fu, Yizhe Chen, and Lin Hua. Thickness-prediction method involving tow redistribution for the dome of composite hydrogen storage vessels. *Polymers*, 14:902, 2022. doi: 10.3390/polym14050902.

[10] Stephen W. Tsai and Edward M. Wu. A general theory of strength for anisotropic materials. *Journal of Composite Materials*, 5:58–80, 1971. doi: 10.1177/002199837100500106.

[11] Z. Hashin. Failure criteria for unidirectional fiber composites. *Journal of Applied Mechanics*, 47:329–334, 1980. doi: 10.1115/1.3153664.

[12] A. Puck and H. Schürmann. Failure analysis of FRP laminates by means of physically based phenomenological models. *Composites Science and Technology*, 58:1045–1067, 1998. doi: 10.1016/S0266-3538(96)00140-6.

[13] S. G. Lekhnitskii. *Theory of Elasticity of an Anisotropic Elastic Body*. Holden-Day, San Francisco, 1963.

[14] Caroline Lüders, Sven Ropte, Daniel Schmidt, and Martin Liebisch. Dataset on hydraulic burst pressure test of type IV composite pressure vessel. *Data in Brief*, 59:111333, 2025. doi: 10.1016/j.dib.2025.111333.

[15] Guido Dhondt. *CalculiX CrunchiX User's Manual, version 2.23*. URL `http://www.dhondt.de/`. Windows build of 15 November 2025 used in this work.

[16] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. arXiv:1412.6980, 2014. URL `https://arxiv.org/abs/1412.6980`.